

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

Register Number	192212293
Name	G Pavan Kumar

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

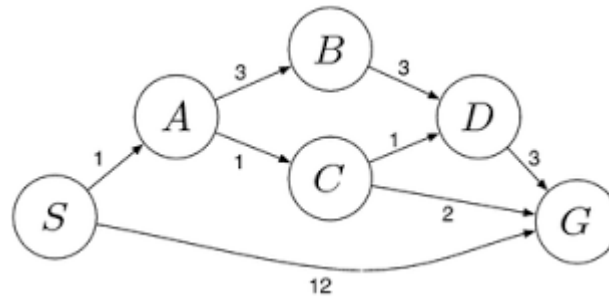
QUESTIONS: 5

Total Marks: 50

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I G Pavan Kumar / (192212293) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shaik Mahaboob Basha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

1. WATER JUG PROBLEM:

Aim:

To solve the Water Jug Problem using a 4-gallon jug and a 3-gallon jug to obtain exactly 2 gallons of water in the 4-gallon jug, without using any measuring marks or other measuring devices.

Problem Statement:

We are given:

- One 4-gallon jug
- One 3-gallon jug
- A water pump

The jugs have no measuring marks. We can:

1. Fill a jug completely from the pump.
2. Empty a jug onto the ground.
3. Pour water from one jug into the other until either the source jug is empty or the destination jug is full.

Objective: Obtain exactly 2 gallons of water in the 4-gallon jug.

Initial State:

4-Gallon Jug = 0 gallons

3-Gallon Jug = 0 gallons

State: (0, 0)

Here, the state is represented as:

(Amount in 4-gallon jug, Amount in 3-gallon jug)

Procedure:

Step	Operation	4-Gallon Jug	3-Gallon Jug	State
1	Fill the 3-gallon jug completely	0	3	(0,3)
2	Pour the 3-gallon jug into the 4-gallon jug	3	0	(3,0)
3	Fill the 3-gallon jug again	3	3	(3,3)
4	Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full	4	2	(4,2)
5	Empty the 4-gallon jug onto the ground	0	2	(0,2)
6	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	2	0	(2,0)

State space representation:

The sequence of states is:

$(0,0) \rightarrow (0,3) \rightarrow (3,0) \rightarrow (3,3) \rightarrow (4,2) \rightarrow (0,2) \rightarrow (2,0)$

Final state:

The final state is:

(2, 0)

Therefore:

- 4-gallon jug = 2 gallons
- 3-gallon jug = 0 gallons

Result:

Thus, exactly 2 gallons of water are obtained in the 4-gallon jug using the given 4-gallon and 3-gallon jugs without any measuring marks or additional measuring devices.

2. MARS ROVER AGENT:

Aim:

To analyze the working of a Mars Rover as an intelligent agent and identify its percepts, operating environment, possible actions, performance measures, and suitable agent architecture.

i. Percepts of Mars rover agent:

The percepts are the information received by the Rover from its environment through sensors and instruments.

The Mars Rover can perceive:

- Images and videos of the Martian surface using cameras.
- Distance and obstacles using sensors.
- Temperature, pressure, and atmospheric conditions.
- The chemical and physical properties of rocks and soil.
- The location and position of the Rover.
- The amount of battery power available.

ii. Characterises Operations of the environment:

The operating environment of a Mars Rover can be characterized as follows:

1. Partially Observable

The Rover cannot observe the entire Martian surface at once. It can only collect information from its sensors and cameras about its nearby surroundings.

2. Deterministic and Stochastic

Some actions may produce predictable results, but unexpected events such as wheel damage, dust storms, or changes in terrain can occur. Therefore, the environment also has uncertain elements.

3. Sequential

The current actions of the Rover affect its future actions. For example, if it uses too much battery while travelling, it may not have enough power to return or complete its mission.

4. Dynamic

The environment may change due to dust storms, temperature changes, and other natural conditions.

5. Continuous

The Rover operates in a continuous environment because its movement, time, temperature, distance, and energy levels can have many possible values.

iii. Actions that agent can take:

The Mars Rover can perform the following actions:

- Move forward, backward, left, or right.
- Change its direction and navigate through different terrains.
- Stop when an obstacle is detected.
- Take photographs and videos.
- Scan rocks and soil.
- Collect soil and rock samples using a robotic arm or drilling system.
- Analyze the collected samples using scientific instruments.
- Identify interesting geological features.
- Avoid obstacles and dangerous areas.

iv. Performance evaluation of the agent:

The performance of the Mars Rover can be evaluated using a performance measure. Some important performance measures are:

- The distance successfully travelled.
- The number of useful areas explored.

- The number and quality of samples collected.
- The accuracy of sample analysis.
- The amount of useful scientific data sent to Earth.
- The successful identification of minerals, rocks, or possible signs of water.
- The ability to avoid obstacles and prevent damage. .

v. Suitable agent of architecture:

The most suitable architecture for a Mars Rover is a Hybrid Agent Architecture, combining a Model-Based Reflex Agent, Goal-Based Agent, and Utility-Based Agent.

Reason

A model-based agent helps the Rover:

- Remember previously explored locations.
- Maintain a map of the surrounding area.
- Track its battery level and equipment condition.
- Understand the effects of previous actions.

A goal-based agent helps the Rover achieve specific goals, such as:

- Reaching a particular location.
- Collecting a sample.
- Returning to a safe location.
- Exploring a selected area.

A utility-based agent helps the Rover select the best action when there are multiple choices. For example, it may choose a route that provides maximum scientific value while using minimum battery power.

Conclusion on Architecture

Therefore, a Hybrid Intelligent Agent is most suitable for a Mars Rover because it can:
Perceive → Understand the environment → Plan → Select the best action → Perform the action → Learn from the result.

3. 8-QUEENS PROBLEM:

Aim:

To formulate the **8-Queens problem** as a search problem and explore the search space using suitable search strategies.

Problem Statement:

The 8-Queens problem requires placing 8 queens on an 8×8 chessboard such that no two queens attack each other. Since a queen can attack horizontally, vertically, and diagonally, no two queens should be placed in the same row, column, or diagonal.

Problem formulation:

The problem can be represented using the following components:

1. Initial State:

An empty chessboard with no queens placed.

2. State:

A state represents a partial arrangement of queens on the chessboard. For example, [1, 5, 8] represents queens placed in the first three columns.

3. Actions:

Place a queen in an empty column in a position where it is not attacked by any previously placed queen.

4. Transition Model:

A new state is generated by placing one queen in the next empty column.

5. Goal Test:

The goal is reached when all 8 queens are placed and no two queens attack each other.

6. Path Cost:

The cost can be considered as the number of moves or queen placements required to reach the solution.

Search Space:

The search space consists of all possible arrangements of 8 queens on the chessboard. If one queen is placed in each column, there are initially:

8^8 possible arrangements

If the condition of one queen per row is also considered, the possibilities are reduced to:

$8! = 40,320$ arrangements

The diagonal constraints further reduce the number of possible states.

Search Strategy:

The most suitable strategy is Depth-First Search (DFS) with Backtracking.

The procedure is:

1. Start with an empty chessboard.
2. Place a queen in the first column.
3. Move to the next column and place a queen in a safe position.
4. Continue placing queens column by column.
5. If no safe position is available, backtrack to the previous column.
6. Move the previous queen to another safe position.
7. Continue until all 8 queens are placed successfully.

Constraints:

For two queens at positions (r_1, c_1) and (r_2, c_2) , they must satisfy:

- $r_1 \neq r_2$ — different rows
- $c_1 \neq c_2$ — different columns
- $|r_1 - r_2| \neq |c_1 - c_2|$ — different diagonals

Result:

Thus, the 8-Queens problem can be formulated as a state-space search problem. The initial state is an empty board, actions place queens in safe positions, and the goal state is a configuration of 8 queens where no two queens attack each other. DFS with backtracking is an efficient search strategy because it eliminates invalid configurations early and backtracks whenever a solution cannot be continued.

4. OLA CAB BOOKING AGENT:**Aim:**

To identify the suitable problem-solving agent for travelling from one location to another using the OLA Cab Booking application and to write the corresponding pseudocode.

Problem statement:

A customer wants to travel from a source location to a destination location using the OLA Cab Booking application. The customer can select a suitable cab type such as Mini, Micro, Sedan, Shared, Prime, etc., based on comfort, cost, and other requirements.

The system must find a suitable cab and route to reach the destination.

Type of problem solving agent:

The most suitable agent is a Utility-Based Agent.

Reason:

A utility-based agent is suitable because the customer has multiple choices and wants to select the best option based on different factors such as:

- Cab type
- Fare
- Travel time
- Distance
- Comfort
- Availability
- Traffic conditions
- Customer preference

The agent evaluates the available choices and selects the cab and route that provide the maximum utility or benefit to the customer.

For example:

- If the customer wants low cost → Shared or Mini cab may be selected.
- If the customer wants more comfort → Prime or Sedan may be selected.
- If the customer wants faster travel → The route with minimum travel time may be selected.

Therefore, the OLA cab booking system can be considered a Utility-Based Problem-Solving Agent.

Problem formulation:

Initial State:

The customer is at the source location.

Goal State:

The customer reaches the destination successfully.

Actions:

The agent can:

- Select the source location.
- Select the destination.
- Display available cab types.
- Check cab availability.
- Calculate fare and travel time.
- Compare available cabs.
- Select the best cab.
- Find the best route.
- Book the cab.
- Track the journey.

Goal Test

The goal is achieved when the customer reaches the selected destination.

Path Cost / Utility

The utility can be calculated based on:

- Minimum fare
- Minimum travel time
- Maximum comfort
- Cab availability
- Customer preferences

Result:

The Utility-Based Agent is the most suitable problem-solving agent for an OLA Cab Booking application. It evaluates different cab types and routes based on cost, comfort, availability, and travel time. It then selects the option that provides the maximum benefit to the customer and helps the customer reach the destination efficiently.

5. UNIFORM COST SEARCH (UCS)

Aim:

To find the least-cost path from the starting warehouse **S** to the destination warehouse **G** using the Uniform Cost Search (UCS) algorithm.

Given weighted graph:

The routes and their costs are:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$

- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Uniform cost search:

Uniform Cost Search always expands the node with the lowest cumulative path cost from the starting node.

The cumulative cost is represented by $g(n)$.

Step by step solution:

Step 1: Start from S

The initial node is:

$S = 0$

Possible paths:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$

Choose **A**, because it has the lowest cost.

Step 2: Expand A

From A:

- $S \rightarrow A \rightarrow B = 1 + 3 = 4$
- $S \rightarrow A \rightarrow C = 1 + 1 = 2$

The lowest cost is **C = 2**.

Step 3: Expand C

From C:

- $S \rightarrow A \rightarrow C \rightarrow D = 2 + 1 = 3$
- $S \rightarrow A \rightarrow C \rightarrow G = 2 + 2 = 4$

The lowest cost is **D = 3**.

Step 4: Expand D

From D:

- $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 3 + 3 = 6$

The available goal paths are:

- Direct path: $S \rightarrow G = 12$
- Via C: $S \rightarrow A \rightarrow C \rightarrow G = 4$
- Via D: $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G = 6$

Since UCS selects the lowest cumulative cost, it selects the path through **C** with cost **4**.

UCS SEARCH TABLE			
	Step	Node Expanded Path	Cumulative Cost
1	S	S	0
2	A	$S \rightarrow A$	1
3	C	$S \rightarrow A \rightarrow C$	2
4	D	$S \rightarrow A \rightarrow C \rightarrow D$	3
5	G	$S \rightarrow A \rightarrow C \rightarrow G$	4

Final Answer:

Least-Cost Path:

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost:

$$1 + 1 + 2 = 4$$

Therefore, using the Uniform Cost Search algorithm, the least-cost path from warehouse **S** to warehouse **G** is:

$S \rightarrow A \rightarrow C \rightarrow G$ with a minimum total cost of 4.

Result:

The UCS algorithm successfully finds the optimal delivery route:

Starting Warehouse $S \rightarrow$ Warehouse $A \rightarrow$ Warehouse $C \rightarrow$ Destination Warehouse G
with a minimum travel cost of 4 units.